LOUIS VINCENT A. CRISALDO

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EDUCATION

Batangas State University – The National Engineering University, Alangilan Campus Alangilan, Batangas City
Bachelor of Science in Computer Science *2022 - Present*

- **General Weighted Average:** 1.53
- **Relevant Coursework:** Database Management System, Data Analysis, Fundamentals of Data Science, Machine Learning

SKILLS

Data Analysis & Programming: Python, Jupyter Notebook, SQL (T-SQL), Power BI (DAX), Microsoft Excel
Project Management & Methodologies: Scrum, OpenProject, Technical Documentation
Tools & Version Control: Playwright, Git, GitHub

EXPERIENCE

Quality Assurance Engineer Intern Batangas State University – TNEU, Pablo Borbon Campus
Center for Artificial Intelligence and Smart Technologies (CAIST) *June 2025 – July 2025*

- Collaborated closely with a multidisciplinary tech team and the senior Lead Developer, streamlining the defect resolution lifecycle via OpenProject to ensure continuous cross-functional alignment.
- Executed comprehensive manual and automated testing (PHPUnit, Playwright), systematically isolating software defects to guarantee a highly reliable system.
- Optimized quality assurance tracking by designing structured defect logs in Microsoft Excel, providing clear, actionable data for the Project Manager and ensuring a seamless technical turnover for newly hired developers.

PROJECTS

Optimization and Domain-Adaptive Pre-Training of DistilmBERT for Sentiment Analysis in Tagalog-English Code-Switched Text
[THESIS]

- Spearheaded a 3-member research team through the end-to-end thesis lifecycle, managing task delegation and research milestones to ensure highly cohesive and on-time task delivery.
- Curated a reliable hybrid training dataset by systematically cleaning and preprocessing unstructured real-world Taglish e-commerce data alongside strategic synthetic data augmentation.
- Executed advanced semantic filtering to ensure high data quality, generating Language Agnostic BERT Sentence (LaBSE) embeddings and applying cosine similarity thresholds to effectively identify and remove near-duplicate entries.
- Engineered a lightweight DistilmBERT architecture for complex Taglish sentiment analysis, shrinking the model footprint from 520MB to 133MB to achieve a 31.42ms inference latency, delivering a 2.98x speedup and 98% performance retention (79.95% macro F1score) compared to the base model.

The Terrain Tax Analysis Pipeline
[PERSONAL PROJECT]

- Engineered a fully automated, end-to-end ETL pipeline utilizing Python and Playwright to systematically harvest, parse, and flatten over 241,000 raw geospatial telemetry logs into a structured tabular format.
- Architected a scalable SQL Server backend environment, deploying optimized T-SQL aggregations and Non-Clustered Indexing (NCI) to process high-variance topographical data.
- Executed a comparative telemetry analysis between Road and Mountain Bike architectures, successfully translating SQL aggregations into a Power BI dashboard to visualize how severe gradient stress neutralizes baseline momentum advantages.

Subscription-Based SaaS Analytics
[PERSONAL PROJECT]

- Engineered a custom SaaS data architecture by synthetically generating over 6,000 raw telemetry logs with realistic anomalies, subsequently building a rigorous Python-based sanitation pipeline to establish a highly reliable analytical dataset.
- Executed complex business intelligence diagnostics utilizing advanced T-SQL (CTEs, Window Functions), successfully isolating a 15.8% "True Churn" rate and identifying immediate retention risks to drive targeted Customer Success interventions.
- Translated relational database metrics into a Power BI executive dashboard via DAX modeling, visualizing MRR concentration and product feature adoption to effectively align strategic decisions with user behavior.